

TITLE PAGE

- Problem Statement ID- **SIH26018**
- Problem Statement Title- *Intelligent Land Record Digitization and Validation System*
- Theme- **Smart Automation**
- PS Category- **Software**
- Team ID-
- Team Name- **PrithviX**



Intelligent Land Record Digitization and Validation System

Current Government Pipeline

Digitization exists, but it is fragmented and largely manual



1. Physical Land Documents

Old, heterogeneous, often handwritten and in multiple formats



2. Scanning / Digitization

Documents scanned and stored as images (PDF/TIFF)



3. Data Entry & Indexing

Manual or semi-manual entry of key details into departmental systems



4. Human Verification / QC

Multiple levels of manual verification (officials at different levels)



5. Digital Land Record

Stored in separate systems (RoR, maps, registration, etc.)

Problems with the Current Pipeline

Digitized ≠ Validated



Heterogeneous Records

Multiple formats, languages, handwritten and degraded documents are difficult to interpret



Manual & Time-Consuming

Heavy reliance on human interpretation and data entry at multiple stages



Fragmented Validation

Textual, spatial and registration data are checked in separate systems with limited integration



Uncertain & Error-Prone

Inconsistencies, missing data and conflicting records are often detected late, leading to delays and disputes

The result: digitized records can still require significant manual effort before they become trusted data.

Our Pipeline (PrithviX)

AI-assisted extraction, validation and integration



Document Upload

Scan or upload land documents (PDF / images)



Content Extraction

Extract text and key information (OCR)



Content Classification

Identify document type and produce structured content (e.g. owner, survey no., area, etc.)



Validate Content

Check accuracy and consistency (rules + GIS verification)



Store in Database

Save validated records (e.g. ULPIN, owner, location, etc.)



Access by Applications

Search, view and use records (GIS, portals, etc.)

How Our System Solves the Problems

Turning challenges into a scalable, reliable solution



Heterogeneous Records



AI-assisted extraction

Handles multiple formats, languages and degraded documents



Manual & Time-Consuming



Automated structured extraction

Reduces manual effort, faster processing



Fragmented Validation



Unified validation engine

Cross-checks textual, historical and spatial data in one place



Uncertain & Error-Prone



Confidence scoring & exception handling

Flags uncertain/conflicting records for human review



Result: higher accuracy, faster processing, reduced workload and more trusted land records.

What is Our Innovation?

A unified, AI-assisted validation layer for land record digitization



EXTRACT
AI-powered document understanding



UNDERSTAND
Structure and normalize key information



CROSS-CHECK
Validate across related records (textual + spatial)



VALIDATE
GIS-based spatial verification



ESCALATE
Confidence-driven human verification with re-validation



Innovation: A modular, end-to-end pipeline that combines document understanding, cross-record consistency and GIS validation with confidence-based human verification — integrated into a single system.

"AI doesn't replace verification — it makes verification exception-driven."

Land Document Digitization & Validation System

From scanned documents to trusted, GIS-verified, ULPIN-linked land records



Modular Microservices
for Scalable & Sustainable
Land Governance

Tech Stack



Frontend

React
(JavaScript)



Backend

Python, Flask



Image Processing

OpenCV



OCR

PaddleOCR /
BharatOCR



AI/NLP

Data Classification
& Extraction



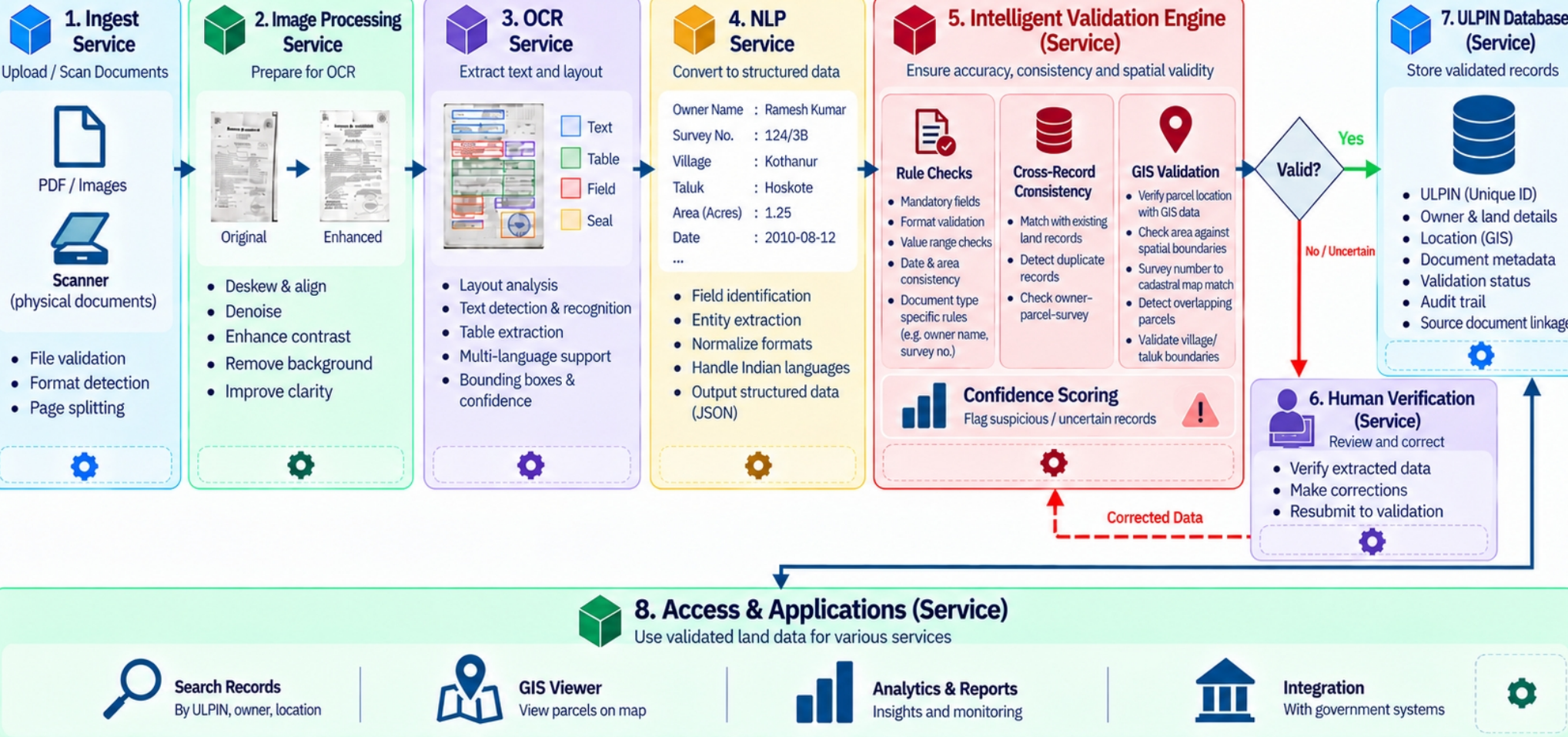
Validation/GIS

Rule Engine,
PostGIS



Database

PostgreSQL +
PostGIS



Analysis of Feasibility

The **proposed system** uses mature technologies for **OCR**, document pre processing, **NLP**, **GIS** and database management. Its **modular micro service** architecture allows ingestion, extraction, classification, validation and storage to be **independently developed**, tested and scaled.

Existing government land-record initiatives such as **DILRMP** provide a practical ecosystem for future integration.

Potential Challenges & Risks

Legacy documents may be **poor-quality** and **damaged**, handwritten, multilingual, or **inconsistently - formatted**, which can reduce extraction accuracy.

Conflicting land information may also make **automated - validation difficult**, while large-scale digitization can increase processing and human-review requirements.

Strategies for Overcoming Challenges

Image pre processing can improve OCR quality, while **AI-based classification** combined with rule based and **GIS validation** can detect inconsistencies. Low confidence records can be routed for **human verification**.

The **modular micro-service architecture** allows individual components to be improved or scaled without redesigning the entire system.



Potential Impact on Target Audience

Accurate, accessible and trustworthy land information for the people who need it



Citizens / Landowners

Simpler access. Greater transparency.

- Online access to verified land records
- Reduced need for physical visits to offices
- Easier dispute checking and property verification
- Builds trust through transparent information

"I can check my land records anytime, without running to offices."

— Citizen



Government Departments

Faster services. Better governance.

- Digitized and standardized records
- Faster verification and decision making
- Early detection of inconsistencies and duplicates
- Reduced manual workload and paperwork
- Seamless integration with DILRMP / ULPIN initiatives

"Reliable, standardized data helps us serve citizens better."

— Government Officer



Surveyors / Field Officials

Accurate information in the field.

- GIS-linked land parcel information
- Quick access to historical and current records
- Helps in field verification and boundary checks
- Digital records reduce paperwork and errors

"Verified information at my fingertips for faster field work."

— Field Official



Financial Institutions

More reliable data for safer lending.

- Verified land information for loan assessment
- Faster processing of property-related loans
- Reduced risk of fraudulent documents
- Greater confidence in land-backed assets

"Accurate land records enable faster and safer lending."

— Bank Representative



Empowered Stakeholders

Transparent Systems

Stronger Land Administration



Benefits of the Solution

From fragmented documents to trusted digital land records



Social Benefits

Greater access. Higher trust.

- Easier access to land records for all citizens, including in rural areas
- Reduced information asymmetry
- Earlier detection of inconsistencies, helping prevent disputes
- More transparent and accountable land governance
- Inclusive access to information for marginalized communities



Economic Benefits

Efficiency and opportunities.

- Faster processing of loans and approvals
- Reduced transaction and administrative costs
- Lower risk of fraud in land transactions
- Improved access to reliable land information for investment and development
- Supports formalization of land assets



Environmental Benefits

Sustainable land use.

- Better land use planning using GIS-enabled data
- Helps prevent encroachment on forests, wetlands and public land
- Enables monitoring of land use changes over time
- Supports climate-resilient and sustainable development
- Facilitates data-driven planning for infrastructure and conservation



Governance Benefits

Data-driven. Accountable. Efficient.

- Centralized and standardized land records
- Faster verification and decision making
- Improved inter-departmental coordination
- Audit trails and source document linkage for accountability
- Supports integration with DILRMP, ULPIN and other government systems
- Enables evidence-based policy and planning



Accurate Records

Transparent Governance

Sustainable Growth

- **Digital India Land Records Modernization Programme (DILRMP)**
dolr.gov.in/en/document-category/program-dilrmp
- **DILRMP MIS 4.0**
dilrmp.gov.in
- **DILRMP 3.0 Operational Guidelines (2026–2031)**
cdnbbsr.s3waas.gov.in/s3d69116f8b0140cdeb1f99a4d5096ffe4/uploads/2026/08/20260827600317271.pdf
- **ULPIN / Bhu-Aadhaar Press release**
www.pib.gov.in/Pressreleaseshare.aspx?PRID=1932867&lang=2®=48
- **Paddle OCR**
www.paddleocr.ai/main/en/index/index.html
- **Open CV**
github.com/opencv/opencv